

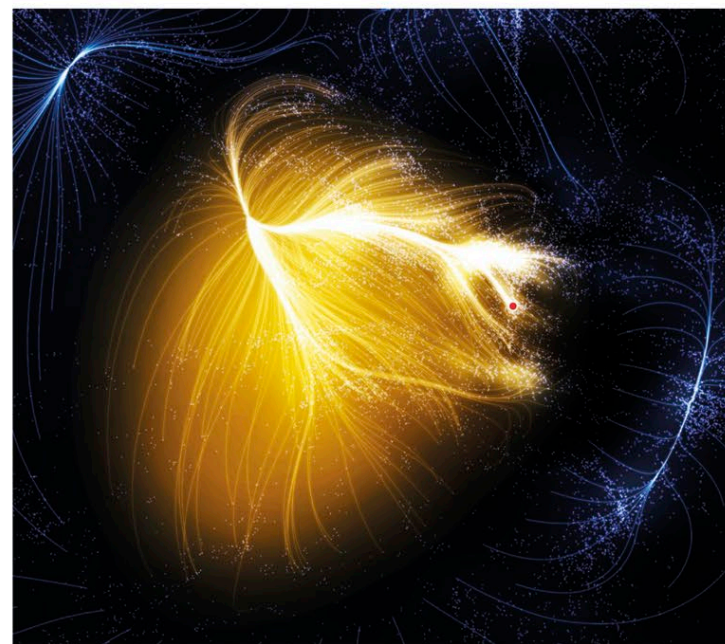
# THE COSMOS

THE UNIVERSE, OR COSMOS, IS EVERYTHING THAT EXISTS—ALL MATTER, ENERGY, TIME, AND SPACE—AND ITS SCALE IS QUITE MIND-BOGGLING. JUST ABOUT EVERYTHING IN IT IS PART OF SOMETHING BIGGER.

The Universe has a hierarchy of structures. Our planet, Earth, is in the Solar System, which lies in the Milky Way Galaxy, itself just one member of a cluster of galaxies called the Local Group. The Local Group in turn is just a part of a larger structure, the Virgo Supercluster. Astronomers have recently identified a vast region of space they have called Laniakea (meaning “immeasurable heaven” in Hawaiian), which contains the Virgo Supercluster and other superclusters. Intriguingly, all the galaxies in it seem to be flowing towards a region at its center, called the Great Attractor.

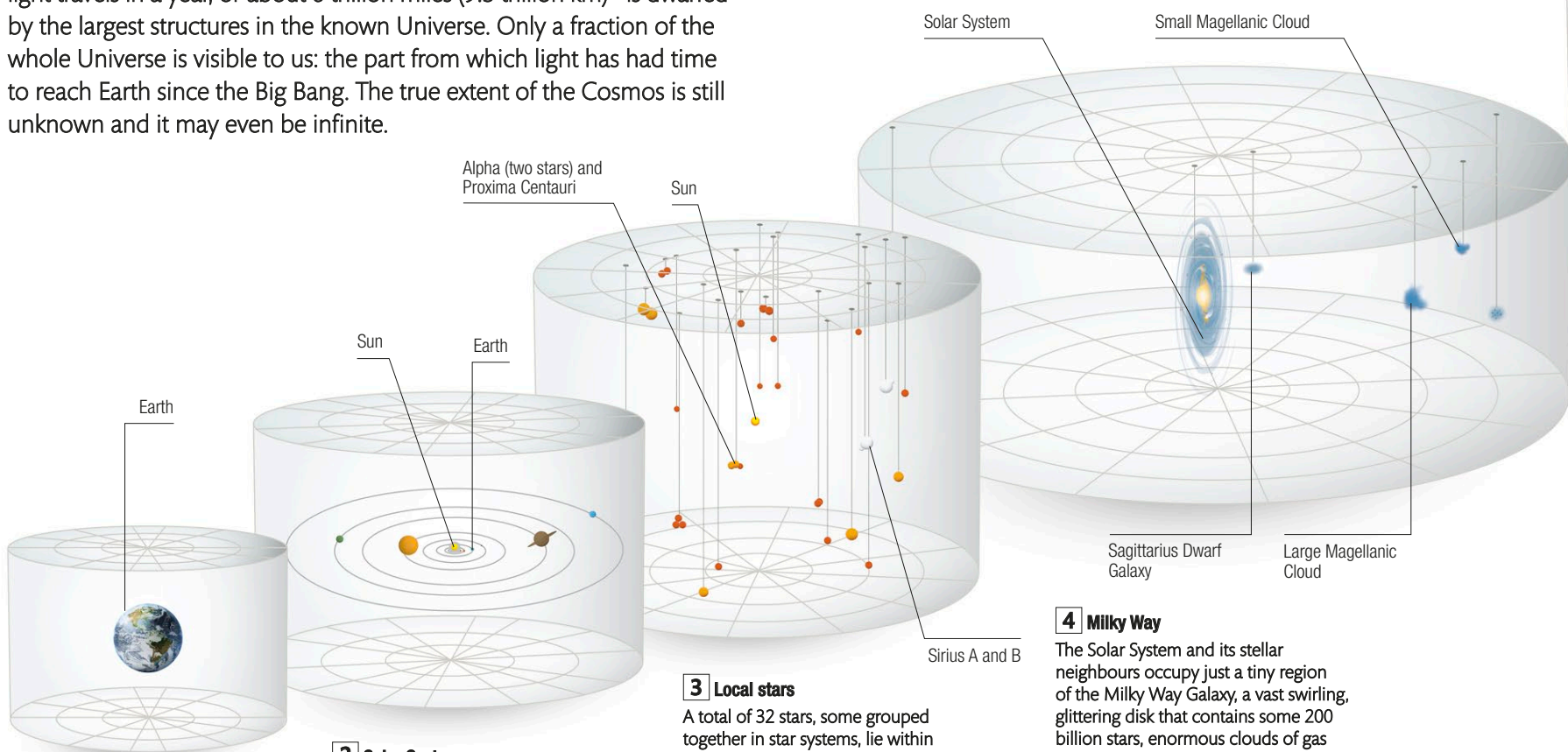
## Light as a yardstick

Astronomers use light as a yardstick for measuring distances because nothing can cross space faster. Yet even one light-year—the distance light travels in a year, or about 6 trillion miles (9.5 trillion km)—is dwarfed by the largest structures in the known Universe. Only a fraction of the whole Universe is visible to us: the part from which light has had time to reach Earth since the Big Bang. The true extent of the Cosmos is still unknown and it may even be infinite.



### △ Laniakea Supercluster

In this depiction of Laniakea (yellow), white lines indicate the flow of galaxies towards a spot near its center. The approximate position of the Milky Way is shown in red. Laniakea is about 500 million light-years across. It is thought to be surrounded by other similar regions (blue).



### 1 Earth

Our home planet is a small rocky sphere floating in the emptiness of space. Earth's closest neighboring object is the Moon. On average, it is a little more than one second away at the speed of light, so one could say that the Moon is one light-second distant.

### 2 Solar System

Earth is part of the Solar System, which comprises our local star, the Sun, and all the objects that orbit the Sun. The most distant planet, Neptune, is about 4.5 hours away at light speed, but the Solar System also includes comets that are up to 1.6 light-years distant.

### 3 Local stars

A total of 32 stars, some grouped together in star systems, lie within 12.5 light-years of the Solar System. They range from dim red dwarfs, invisible to the naked eye, to dazzling, yellow or white, Sun-like stars. A few are suspected to have their own planets.

### 4 Milky Way

The Solar System and its stellar neighbours occupy just a tiny region of the Milky Way Galaxy, a vast swirling, glittering disk that contains some 200 billion stars, enormous clouds of gas and dust, and a supermassive black hole at its center. The Milky Way is over 100,000 light-years across. Surrounding it are several smaller, satellite galaxies.